PassFrame tuition

Number Codes

Year 5 Accelerated — Homework ⏱ 45 min

Worked example — read this through before you start

Worked example: If the code for 45 is 54 (the digits are reversed), what is the code for 78?

Method & answer: Reverse the digits of 78: 87. Answer: 87.

Easy / Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. If the code for 17 is 71 (the digits are reversed), what is the code for 29?

2. If the code for 38 is 83 (the digits are reversed), what is the code for 52?

3. If each digit shifts forward by 1 (wrapping 9 back round to 0), 12 becomes 23. What is the code for 34?

4. If each digit shifts forward by 2 (wrapping 9 back round to 0), 21 becomes 43. What is the code for 56?

5. If each digit shifts back by 1 (wrapping 0 back round to 9), 46 becomes 35. What is the code for 78?

6. If the code for 15 is 51 (the digits are reversed), what is the code for 37?

7. If each digit shifts forward by 1 (wrapping 9 back round to 0), 203 becomes 314. What is the code for 122?

8. If the code for 16 is 7 (the digits are added together), what is the code for 28?

9. If the code for 24 is 6 (the digits are added together), what is the code for 53?

10. If each digit shifts forward by 2 (wrapping 9 back round to 0), 234 becomes 456. What is the code for 111?

11. If the code for 28 is 82 (the digits are reversed), what is the code for 64?

12. If each digit shifts back by 3 (wrapping 0 back round to 9), 68 becomes 35. What is the code for 95?

13. If the code for 35 is 8 (the digits are added together), what is the code for 47?

14. If 7 # 4 = 11 (the rule is: add the two numbers), what is 9 # 6?

15. If 10 # 3 = 7 (the rule is: subtract the second number from the first), what is 12 # 5?

Standard — *work through together, answer 5 in the lesson* ⏱ *15 min (1 min per question)*

1. If each digit shifts forward by 4 (wrapping 9 back round to 0), 3187 becomes 7521. What is the code for 5629?

2. If the code for 2468 is 8642 (the digits are reversed), what is the code for 9135?

3. If the code for 357 is 15 (the digits are added together), what is the code for 648?

6. If 12 # 3 = 30 (the rule is: multiply the two numbers, then subtract the second number's double), what is 15 # 6?

7. If the code for 5093 is 3905 (the digits are reversed), what is the code for 7264?

8. If each digit shifts forward by 5 (wrapping 9 back round to 0), 276 becomes 721. What is the code for

11. If each digit shifts back by 4 (wrapping 0 back round to 9), 918 becomes 574. What is the code for 605?

12. If 6 # 7 = 26 (the rule is: add the two numbers, then double the result), what is 14 # 11?

13. If the code for 6710 is 0176 (the digits are reversed), what is the code for 4529?

813?

4. If each digit shifts back by 3 (wrapping 0 back round to 9), 7051 becomes 4728. What is the code for 4826?

5. If 14 # 6 = 34 (the rule is: double the first number, then add the second), what is 20 # 9?

9. If the code for VOYAGE is 75 (add up the alphabet positions: V=22, O=15, Y=25, A=1, G=7, E=5, so 22+15+25+1+7+5=75), what is the code for ISLAND using the same rule?

10. If the code for 583 is 16 (the digits are added together), what is the code for 726?

14. If each digit shifts forward by 4 (wrapping 9 back round to 0), 5934 becomes 9378. What is the code for 1287?

15. If the code for MEADOW is 61 (add up the alphabet positions: M=13, E=5, A=1, D=4, O=15, W=23, so 13+5+1+4+15+23=61), what is the code for CANYON using the same rule?

Standard / Advanced — *work through together, answer 5 in the lesson* *15 min (1 min per question)*

1. If 23 codes as 56, what codes for 45?

2. If 23 codes as 15, what codes for 52?

3. If 12 codes as 24, what codes for 13?

4. If 12 codes as 32, what codes for 34?

5. If 3 ~ 5 = 15 and 8 ~ 2 = 6, what is

6. A code adds 1 to the first digit, 2 to the second digit, and 3 to the third digit of a 3-digit number, wrapping round past 9 back to 0. Applying this to 213: 2+1=3, 1+2=3, 3+3=6, giving 336. What does the same rule turn 578 into?

7. A code reverses the digits of a number, then adds the original number's digit sum to the result. Applying this to 34: reverse to 43, digit sum of 34 is 7, so 43+7=50. What does the same rule turn 27 into?

8. A code works by multiplying the two digits of a 2-digit number and then adding the original number. Applying this to 23: 2×3=6, then 23+6=29. What does the same rule turn 34 into?

9. In a code, the first digit of a 3-digit number shifts forward by the value of the last digit (the other digits stay the same). Applying this to 253: the last digit is 3, so the first digit 2 becomes 2+3=5, giving 553. What does the same rule turn 174 into?

10. A code takes a 3-digit number,

11. If the code for CAT is 3120 (write down each letter's alphabet position and run the numbers together: C=3, A=1, T=20, giving 3-1-20 → 3120), what is the code for DOG?

12. A code doubles a 2-digit number and then reverses the digits of the result. Applying this to 14: double to 28, then reverse to get 82. What does the same rule turn 23 into?

13. A code adds the two digits of a number together to make a new tens digit, and keeps the original units digit, forming a new 2-digit number. Applying this to 27: 2+7=9 (new tens digit), the units digit stays 7, giving 97. What does the same rule turn 35 into?

14. A code raises the first digit of a 2-digit number to the power of the second digit. Applying this to 23: $2^3 = 8$. What does the same rule turn 32 into?

15. A code subtracts the reverse of

4 ~ 7?

reverses it, and then subtracts the middle digit of the ORIGINAL number from the reversed number. Applying this to 412: reverse to 214, the middle digit of 412 is 1, so 214−1=213. What does the same rule turn 326 into?

a number from double the number. Applying this to 24: double is 48, the reverse of 24 is 42, so 48−42=6. What does the same rule turn 35 into?